Characterization of Message Lengths for Capacity-Achieving Linear PIR

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***Abstract*—We characterize all positive message lengths of capacity-achieving linear private information retrieval (CALPIR) schemes with $M \geq 2$ replicated messages and $N \geq 2$ noncolluding servers. Over every finite field, the achievable lengths are precisely $a(N-1)+bN^{M-1}$, where a, b are nonnegative integers, not both zero. Necessity follows from the pointwise equality conditions of the PIR capacity converse and the integrality of linear conditional entropies. We also revisit T-private PIR for $2 \leq T < N$ and give a pointwise entropy proof that every capacity-achieving linear scheme satisfies $\gcd(N,T)(N/\gcd(N,T))^{M-1} \mid L$. Known constructions attain every positive multiple of this quantity over fields of cardinality at least N. The T-private divisibility threshold and large-field constructions are known from earlier subpacketization work; our treatment gives an explicit querywise entropy derivation under the stated download convention. We interpret the different length spectra through querywise rank integrality, server symmetry, and the distribution of single-server information.**

I. Problem Setting and Related Work

Let $M, N \geq 2$ and $L \geq 1$ be integers, and let $\mathbb{F}_p$ be a finite field of cardinality p, where p is a prime power. We put $[M] := \{1, \ldots, M\}$ and $[N] := \{1, \ldots, N\}$. All entropies use logarithms to base p. The M messages $W^1, \ldots, W^M \in \mathbb{F}_p^L$ are independent and uniformly distributed, and are replicated at all N servers. The user selects an index θ, uniformly distributed on $[M]$, and draws queries $Q_n \in \mathcal{Q}_n$ for $n \in [N]$, where each query alphabet $\mathcal{Q}_n$ is finite. The answer of server n is A_n. Write $Q_{[N]} = (Q_1, \ldots, Q_N)$, $A_{[N]} = (A_1, \ldots, A_N)$, and $W^{[M]} = ((W^1)^\top, \ldots, (W^M)^\top)^\top$. The requirements are

1) $\mathrm{I}(\theta, Q_{[N]}; W^{[M]}) = 0$ (query independence);
2) $\mathrm{H}(A_n \mid Q_n, W^{[M]}) = 0$ for each n (deterministic answers);
3) $\mathrm{H}(W^m \mid A_{[N]}, Q_{[N]}, \theta = m) = 0$ for every $m \in [M]$ (zero-error correctness);
4) $\Pr(Q_n = q \mid \theta = m) = \Pr(Q_n = q \mid \theta = m')$ for every n, m, m', q (one-server privacy).

A scheme is *linear* if, for every n and $q \in \mathcal{Q}_n$, there is a matrix $\mathbf{G}_n^q$ over $\mathbb{F}_p$ such that $A_n = \mathbf{G}_n^{Q_n} W^{[M]}$. The number of rows of $\mathbf{G}_n^q$ may depend on q; thus the answer is a vector of query-dependent length. We measure the expected *entropy download* by

$$D := \sum_{n \in [N]} \mathrm{H}(A_n \mid Q_n). \tag{1}$$

For a fixed query, deleting linearly dependent answer coordinates preserves the information content and recoverability. Accordingly, in the linear setting D can also be implemented as the expected number of transmitted, linearly independent field symbols, with the query-dependent choice of a row-space basis known to the user. Without this reduction, the actual number of transmitted coordinates may exceed D.

The PIR capacity converse [1] states that

$$\frac{D}{L} \geq \sum_{j=0}^{M-1} N^{-j} = \frac{1 - N^{-M}}{1 - N^{-1}}. \tag{2}$$

We call a linear scheme attaining equality *CALPIR*. Sun and Jafar constructed capacity-achieving schemes with $L = N^{M-1}$ [3]; Tian, Sun, and Chen constructed such schemes with $L = N - 1$ and established the minimum linear message length [2]. Our question is not merely the minimum: we determine *every* positive L for which equality in (2) is possible.

For colluding servers, Zhang and Xu studied the smallest subpacketization of linear capacity-achieving T-private PIR and established a threshold $\gcd(N,T)(N/\gcd(N,T))^{M-1}$ [5]; divisibility arguments already appear in their proof. Xu and Zhang subsequently constructed schemes at the threshold over fields with cardinality at least N [6]. We explicitly allow query-dependent answer dimensions and use the conditional-entropy download convention. The pointwise proof below works directly in this formulation; we do not claim that this distinction alone establishes a new achievable-length result beyond the earlier subpacketization theory.

II. Capacity Converse and Its Equality Conditions

We give the capacity converse in detail, following the sequence of entropy manipulations: recovery, restriction to T answers, T-privacy, Han's inequality, and recovery of the next message. The equality conditions at each of these operations will be needed.

For $S \subseteq [M]$, the symbol W^S denotes the collection of messages whose indices belong to S. The following elementary locality identity will be used throughout. For every supported $\mathbf{q}$ under $\theta = m$ and every $\mathcal{T} \subseteq [N]$,

$$\begin{aligned} &\mathrm{H}(A_{\mathcal{T}} \mid Q_{[N]} = \mathbf{q}, W^S, \theta = m) \\ &\quad = \mathrm{H}(A_{\mathcal{T}} \mid Q_{\mathcal{T}} = q_{\mathcal{T}}, W^S, \theta = m). \end{aligned} \tag{3}$$

Indeed, the answers in $\mathcal{T}$ are deterministic functions of $(Q_{\mathcal{T}}, W^{[M]})$, and the conditional distribution of $W^{[M]}$ does

not depend on $(\theta, Q_{[N]})$. Likewise, for fixed $q_{\mathcal{T}}$ the right-hand side of (3) is independent of the requested index.

Lemma 1 (Equality in Han's inequality)**.** *Let $X_1, \dots, X_N, Z$ be discrete random variables with finite conditional entropies and let $1 \le T < N$. If*

$$\frac{1}{\binom{N}{T}} \sum_{|\mathcal{T}|=T} \mathrm{H}(X_{\mathcal{T}} \mid Z) = \frac{T}{N} \mathrm{H}(X_{[N]} \mid Z), \tag{4}$$

then $X_1, \dots, X_N$ are mutually conditionally independent given Z.

Proof. For $0 \le k \le N$, set $F_0 = 0$ and

$$F_k = \frac{1}{\binom{N}{k}} \sum_{|\mathcal{T}|=k} \mathrm{H}(X_{\mathcal{T}} \mid Z) \quad (k \ge 1).$$

Let $\Delta_k = F_k - F_{k-1}$. The chain rule expresses Δ_k as the average of $\mathrm{H}(X_i \mid X_{\mathcal{T}}, Z)$ over all pairs $(\mathcal{T}, i)$ with $|\mathcal{T}| = k-1$ and $i \notin \mathcal{T}$. To compare Δ_k and Δ_{k+1}, choose $\mathcal{T}$ uniformly among $(k-1)$-subsets, then choose distinct i, j uniformly outside $\mathcal{T}$. The first increment is the expectation of $\mathrm{H}(X_i \mid X_{\mathcal{T}}, Z)$, and the second is that of $\mathrm{H}(X_i \mid X_{\mathcal{T}}, X_j, Z)$; therefore

$$\Delta_k - \Delta_{k+1} = \mathbb{E}\big[\mathrm{I}(X_i; X_j \mid X_{\mathcal{T}}, Z)\big] \ge 0.$$

Thus $\Delta_1 \ge \cdots \ge \Delta_N$. The assumed equality says $F_T/T = F_N/N$. Since $T < N$, the average of the first T terms of a nonincreasing sequence equals the average of all N terms only if all terms are equal. Hence

$$\sum_{n \in [N]} \mathrm{H}(X_n \mid Z) = N\Delta_1 = F_N = \mathrm{H}(X_{[N]} \mid Z).$$

Equality in conditional entropy subadditivity gives the desired conditional independence. □

Lemma 2 (Pointwise equalities)**.** *Suppose $1 \le T < N$ and T-privacy holds. If the scheme attains the T-private capacity, then for every supported query $\mathbf{q}$ under $\theta = m$ and every nonempty $J \subseteq [M]$:*

1) *If $m \in J$, then*

$$\mathrm{H}(A_{[N]} \mid Q_{[N]} = \mathbf{q}, W^{[M]\setminus J}, \theta = m) = \sum_{n \in [N]} \mathrm{H}(A_n \mid Q_n = q_n, W^{[M]\setminus J}, \theta = m). \tag{5}$$

2) *If $m \notin J$, for each T-subset $\mathcal{T}$,*

$$\mathrm{H}(A_{[N]} \mid Q_{[N]} = \mathbf{q}, W^{[M]\setminus J}, \theta = m) = \mathrm{H}(A_{\mathcal{T}} \mid Q_{\mathcal{T}} = q_{\mathcal{T}}, W^{[M]\setminus J}, \theta = m). \tag{6}$$

3) *If $m \in J$, then independently of capacity achievement,*

$$\mathrm{H}(A_{[N]} \mid Q_{[N]} = \mathbf{q}, W^{[M]\setminus J}, \theta = m) = L + \mathrm{H}(A_{[N]} \mid Q_{[N]} = \mathbf{q}, W^{[M]\setminus (J\setminus\{m\})}, \theta = m). \tag{7}$$

Proof. Step 1: The complete capacity converse. Fix any permutation $m_1, \dots, m_M$ of $[M]$ and put $S_j := \{m_1, \dots, m_j\}$ for $1 \le j \le M$. Since privacy makes the local answer entropy independent of the requested index, we begin with

$$\begin{aligned} D &= \sum_{n \in [N]} \mathrm{H}(A_n \mid Q_n, \theta = m_1) \\ &= \sum_{n \in [N]} \mathrm{H}(A_n \mid Q_{[N]}, \theta = m_1) \\ &\ge \mathrm{H}(A_{[N]} \mid Q_{[N]}, \theta = m_1) \\ &= L + \mathrm{H}(A_{[N]} \mid Q_{[N]}, W^{m_1}, \theta = m_1). \end{aligned} \tag{8}$$

Here the second line uses (3), the inequality is entropy subadditivity, and the final equality is the chain rule: correctness gives $\mathrm{H}(W^{m_1} \mid A_{[N]}, Q_{[N]}, \theta = m_1) = 0$, whereas query independence gives $\mathrm{H}(W^{m_1} \mid Q_{[N]}, \theta = m_1) = L$.

For each $j = 1, \dots, M-1$, restrict the answers to a T-subset, change the requested index, and apply Han's inequality:

$$\begin{aligned} &\mathrm{H}(A_{[N]} \mid Q_{[N]}, W^{S_j}, \theta = m_j) \\ &\ge \frac{1}{\binom{N}{T}} \sum_{|\mathcal{T}|=T} \mathrm{H}(A_{\mathcal{T}} \mid Q_{\mathcal{T}}, W^{S_j}, \theta = m_j) \\ &= \frac{1}{\binom{N}{T}} \sum_{|\mathcal{T}|=T} \mathrm{H}(A_{\mathcal{T}} \mid Q_{\mathcal{T}}, W^{S_j}, \theta = m_{j+1}) \\ &\ge \frac{T}{N} \mathrm{H}(A_{[N]} \mid Q_{[N]}, W^{S_j}, \theta = m_{j+1}). \end{aligned} \tag{9}$$

The first inequality holds for every subset separately because an answer subset cannot have more entropy than the full answers. The middle equality holds by T-privacy and query–message independence: the distribution of $(Q_{\mathcal{T}}, W^{[M]})$ is the same for both desired indices. For the last inequality, first condition on the complete query, apply Han's inequality to the N answers, and then use (3) for every T-subset.

The next message $W^{m_{j+1}}$ can be recovered from the full answers at desired index m_{j+1}. The chain rule therefore gives

$$\begin{aligned} &\mathrm{H}(A_{[N]} \mid Q_{[N]}, W^{S_j}, \theta = m_{j+1}) \\ &= \mathrm{H}(W^{m_{j+1}}, A_{[N]} \mid Q_{[N]}, W^{S_j}, \theta = m_{j+1}) \\ &= L + \mathrm{H}(A_{[N]} \mid Q_{[N]}, W^{S_{j+1}}, \theta = m_{j+1}). \end{aligned} \tag{10}$$

The first equality uses correctness even after revealing W^{S_j}; the second uses independence of $W^{m_{j+1}}$ from $(W^{S_j}, Q_{[N]}, \theta)$. Combining (9) and (10) gives the recursion

$$\mathrm{H}(A_{[N]} \mid Q_{[N]}, W^{S_j}, \theta = m_j) \ge \frac{T}{N}\big[L + \mathrm{H}(A_{[N]} \mid Q_{[N]}, W^{S_{j+1}}, \theta = m_{j+1})\big]. \tag{11}$$

For $j = M-1$, the remaining entropy after revealing $S_M = [M]$ is zero by answer determinism. Substituting the recursion

into (8), one stage at a time, gives

$$\begin{aligned} D &\geq L + \mathrm{H}(A_{[N]} \mid Q_{[N]}, W^{S_1}, \theta = m_1) \\ &\geq L + \frac{T}{N}\Big[L + \mathrm{H}(A_{[N]} \mid Q_{[N]}, W^{S_2}, \theta = m_2)\Big] \\ &\geq L + \frac{T}{N}L + \left(\frac{T}{N}\right)^2 \Big[L + \cdots\Big] \\ &= L \sum_{j=0}^{M-1} \left(\frac{T}{N}\right)^j. \end{aligned}$$

Thus the standard T-private capacity converse follows without suppressing the intermediate entropy inequalities.

Step 2: Equality in every intermediate inequality. At capacity, the first and last terms of the preceding chain coincide. Every omitted slack is nonnegative and has a strictly positive coefficient, since $T/N > 0$. Consequently the initial subadditivity inequality in (8) and *both* inequalities in (9) must be equalities. This conclusion holds for *every* permutation. In particular, given $\emptyset \neq S \subsetneq [M]$, $m \in S$, and $m' \notin S$, choose a permutation with S as its first $|S|$ indices, with m last in S and m' next. Then both inequalities in (9), with these S, m, m', are equalities.

Step 3: From average equalities to every supported query. Fix S, m as above and $|\mathcal{T}| = T$. For each $\mathbf{q}$ supported under $\theta = m$, locality and entropy monotonicity give

$$\begin{aligned} &\mathrm{H}(A_{[N]} \mid Q_{[N]} = \mathbf{q}, W^S, \theta = m) \\ &\quad - \mathrm{H}(A_{\mathcal{T}} \mid Q_{\mathcal{T}} = q_{\mathcal{T}}, W^S, \theta = m) \geq 0. \end{aligned}$$

The expectation of the arithmetic average of these gaps over all T-subsets is exactly the gap in the first inequality of (9), which is zero by Step 2. Each summand is nonnegative and there are finitely many T-subsets and supported queries. Therefore *every* individual gap vanishes at every supported $\mathbf{q}$.

For a supported $\mathbf{q}$ under $\theta = m'$, the conditional Han gap is also nonnegative:

$$\begin{aligned} &\frac{1}{\binom{N}{T}} \sum_{|\mathcal{T}|=T} \mathrm{H}(A_{\mathcal{T}} \mid Q_{\mathcal{T}} = q_{\mathcal{T}}, W^S, \theta = m') \\ &\quad - \frac{T}{N}\mathrm{H}(A_{[N]} \mid Q_{[N]} = \mathbf{q}, W^S, \theta = m') \geq 0. \end{aligned}$$

Its expectation is zero by Step 2, so it vanishes pointwise. Lemma 1, applied after fixing $\mathbf{q}$ and conditioning on W^S, now shows that the answers are mutually conditionally independent for that supported query.

Step 4: Identification of the three claims. For $m \notin J$, put $S = [M] \setminus J$; then $m \in S$ and the first gap from Step 3 gives (6). For $m \in J$ with $J \neq [M]$, again put $S = [M] \setminus J$; now $m \notin S$, and the pointwise Han equality gives (5). For $J = [M]$, use the initial subadditivity equality with $m_1 = m$, lifted pointwise by the same nonnegative-gap argument.

Finally let $m \in J$, put $S = [M] \setminus J$, and fix a supported $\mathbf{q}$. The chain rule and correctness give

$$\begin{aligned} &\mathrm{H}(A_{[N]} \mid Q_{[N]} = \mathbf{q}, W^S, \theta = m) \\ &= \mathrm{H}(W^m, A_{[N]} \mid Q_{[N]} = \mathbf{q}, W^S, \theta = m) \\ &= L + \mathrm{H}(A_{[N]} \mid Q_{[N]} = \mathbf{q}, W^{S\cup\{m\}}, \theta = m), \end{aligned} \tag{12}$$

where the first equality uses zero-error recovery and the second uses independence of the messages and queries. Since $S \cup \{m\} = [M]\setminus(J\setminus\{m\})$, this is precisely (7). This final equality does not require capacity achievement. □

III. Complete Classification for Noncolluding Servers

Define the positive part of the numerical semigroup

$$\mathrm{AM} := \{a(N-1) + bN^{M-1} : a, b \in \mathbb{Z}_{\geq 0}\} \setminus \{0\}. \tag{13}$$

Theorem 1. *For any finite field $\mathbb{F}_p$ and any integers $M, N \geq 2$, a CALPIR scheme with positive message length L exists if and only if $L \in \mathrm{AM}$.*

Proof. The constructions with message lengths $N-1$ and N^{M-1} exist over every finite field [2], [3]. Direct sums preserve linearity, correctness, privacy, and equality in the normalized download bound, proving sufficiency.

For necessity fix $m \in [M]$ and a supported query vector $\mathbf{q}$. Write

$$h(\mathbf{q}) := \mathrm{H}(A_{[N]} \mid Q_{[N]} = \mathbf{q}, W^m, \theta = m).$$

Take $T = 1$. By Lemma 2 with $J = [M]$ and $J = [M]\setminus\{m\}$, respectively, and by correctness,

$$\begin{aligned} Nh(\mathbf{q}) &= \sum_{n\in[N]} \mathrm{H}(A_n \mid Q_{[N]} = \mathbf{q}, W^m, \theta = m) \\ &\leq \sum_{n\in[N]} \mathrm{H}(A_n \mid Q_{[N]} = \mathbf{q}, \theta = m) \\ &= \mathrm{H}(A_{[N]} \mid Q_{[N]} = \mathbf{q}, \theta = m) \\ &= L + h(\mathbf{q}). \end{aligned} \tag{14}$$

The last identity is the chain rule using the zero-error recovery of W^m and query independence. Since the scheme is linear, $h(\mathbf{q})$ is the rank of the answer map restricted to the $M-1$ undesired message blocks and hence is an integer. Thus

$$h(\mathbf{q}) \leq \left\lfloor \frac{L}{N-1} \right\rfloor. \tag{15}$$

Averaging over supported queries and using the equality $D = \mathrm{H}(A_{[N]} \mid Q_{[N]}, \theta = m)$ from the converse,

$$\begin{aligned} D - L &= \mathrm{H}(A_{[N]} \mid Q_{[N]}, W^m, \theta = m) \\ &\leq \left\lfloor \frac{L}{N-1} \right\rfloor. \end{aligned} \tag{16}$$

Capacity attainment and (2) give

$$L - \frac{L}{N^{M-1}} \leq (N-1)\left\lfloor \frac{L}{N-1} \right\rfloor. \tag{17}$$

Let b be the least nonnegative residue of L modulo $N-1$. Then (17) is equivalent to $bN^{M-1} \le L$. Since $N^{M-1} \equiv 1 \pmod{N-1}$,

$$a := \frac{L - bN^{M-1}}{N-1} \in \mathbb{Z}_{\ge 0}.$$

Consequently $L = a(N-1) + bN^{M-1} \in \mathrm{AM}$. □

A. Consequences of the Classification

For $N = 2$, $\mathrm{AM} = \mathbb{Z}_{\ge 1}$. If $N \ge 3$, the two semigroup generators are coprime. The Frobenius coin theorem then shows that the largest positive length outside AM is

$$(N-2)(N^{M-1}-1) - 1. \tag{18}$$

This formula also holds when $(M,N) = (2,3)$; only the intermediate interval considered below is empty in that case. There are exactly

$$\frac{(N-2)(N^{M-1}-1)}{2} \tag{19}$$

nonachievable positive lengths, and every $L \ge (N-2)(N^{M-1}-1)$ is achievable. For $N \ge 3$ and $(M,N) \ne (2,3)$, the number of achievable lengths in $[N-1, (N-2)(N^{M-1}-1)-1]$ is

$$\frac{(N-2)(N^{M-1}-1)}{2} - 1. \tag{20}$$

For $(M,N) = (2,3)$, $\mathrm{AM} = \mathbb{Z}_{\ge 2}$.

IV. Extension to T-Private PIR

Suppose $2 \le T < N$. Privacy is now required for every $\mathcal{T} \subseteq [N]$ with $|\mathcal{T}| \le T$:

$$\begin{aligned} &\Pr(Q_{\mathcal{T}} = q_{\mathcal{T}} \mid \theta = m) \\ &\quad = \Pr(Q_{\mathcal{T}} = q_{\mathcal{T}} \mid \theta = m') \end{aligned} \tag{21}$$

for all $m, m' \in [M]$ and all $q_{\mathcal{T}}$. The capacity bound is [4]

$$\frac{D}{L} \ge \sum_{j=0}^{M-1} (T/N)^j. \tag{22}$$

Theorem 2. *Every capacity-achieving linear T-private PIR scheme over any finite field $\mathbb{F}_p$ has a message length satisfying*

$$\gcd(N,T) \left(\frac{N}{\gcd(N,T)} \right)^{M-1} \mid L. \tag{23}$$

Conversely, for every $p \ge N$ and every positive L satisfying (23)*, such a scheme exists over $\mathbb{F}_p$.*

The numerical threshold in (23) and the large-field construction are known from [5], [6]. The argument below proves the divisibility condition directly with pointwise conditional entropy equalities, allowing the answer dimension to depend on the query. No distinct novelty of the numerical threshold is asserted.

Lemma 3 (Querywise server symmetry)**.** *Fix a nonempty proper set $J \subsetneq [M]$. For every $m \in [M]$ and every query $\mathbf{q}$ supported under $\theta = m$, the numbers*

$$\mathrm{H}(A_n \mid Q_n = q_n, W^{[M]\setminus J}, \theta = m), \quad n \in [N], \tag{24}$$

are all equal.

Proof. Write $r_n(q_n) := \mathrm{H}(A_n \mid Q_n = q_n, W^{[M]\setminus J})$. By (3), $r_n(q_n)$ is independent of the desired index as long as q_n is fixed. Choose $m' \in J$ and $m'' \notin J$.

Case 1: A query supported for m''. Fix *one* full query $\mathbf{q}'$ supported under $\theta = m''$. For each T-subset $\mathcal{T}$, privacy implies that

$$\Pr(Q_{\mathcal{T}} = q'_{\mathcal{T}} \mid \theta = m') = \Pr(Q_{\mathcal{T}} = q'_{\mathcal{T}} \mid \theta = m'') > 0.$$

Hence there exists a full query $\mathbf{q}^{\mathcal{T}}$ supported under $\theta = m'$ such that $q^{\mathcal{T}}_{\mathcal{T}} = q'_{\mathcal{T}}$. With the *same fixed* $\mathbf{q}'$ on the right throughout,

$$\begin{aligned} &\sum_{n \in \mathcal{T}} r_n(q'_n) = \sum_{n \in \mathcal{T}} r_n(q^{\mathcal{T}}_n) \\ &= \mathrm{H}(A_{\mathcal{T}} \mid Q_{\mathcal{T}} = q^{\mathcal{T}}_{\mathcal{T}}, W^{[M]\setminus J}, \theta = m') \\ &= \mathrm{H}(A_{\mathcal{T}} \mid Q_{\mathcal{T}} = q'_{\mathcal{T}}, W^{[M]\setminus J}, \theta = m'') \\ &= \mathrm{H}(A_{[N]} \mid Q_{[N]} = \mathbf{q}', W^{[M]\setminus J}, \theta = m''). \end{aligned} \tag{25}$$

The second equality is (5) at $\mathbf{q}^{\mathcal{T}}$; the third uses privacy and query independence, and the last is (6) at $\mathbf{q}'$. The right-hand side does not depend on $\mathcal{T}$. For distinct n, n', choose a subset of size $T-1$ disjoint from $\{n, n'\}$ and compare its unions with $\{n\}$ and $\{n'\}$. Since $T < N$, such a subset exists. Thus $r_n(q'_n) = r_{n'}(q'_{n'})$.

Case 2: An arbitrary supported query. Let $\mathbf{q}$ be supported for any desired index m and let $n \ne n'$. Since $T \ge 2$, choose a T-subset containing both n, n'. Privacy gives a full query $\mathbf{q}'$ supported for $\theta = m''$ with $q'_{\mathcal{T}} = q_{\mathcal{T}}$. By Case 1,

$$r_n(q_n) = r_n(q'_n) = r_{n'}(q'_{n'}) = r_{n'}(q_{n'}).$$

This proves the assertion for every supported full query. □

Lemma 4 (One-step entropy recurrence)**.** *Let $J \subseteq [M]$ with $2 \le |J| \le M-1$, choose $m \in J$, and fix a query $\mathbf{q}$ supported under $\theta = m$. Choose $m' \in J \setminus \{m\}$ and any T-subset $\mathcal{T}$. Then some full query $\mathbf{q}'$ supported under $\theta = m'$ satisfies $q'_{\mathcal{T}} = q_{\mathcal{T}}$, and for any $n' \in [N]$,*

$$\begin{aligned} &\sum_{n \in [N]} \mathrm{H}(A_n \mid Q_n = q_n, W^{[M]\setminus J}, \theta = m) \\ &= L + T\mathrm{H}(A_{n'} \mid Q_{n'} = q'_{n'}, \\ &\qquad W^{[M]\setminus (J \setminus \{m\})}, \theta = m'). \end{aligned} \tag{26}$$

Proof. The existence of $\mathbf{q}'$ follows from T-privacy: the T-tuple $q_{\mathcal{T}}$ has positive probability at $\theta = m$, hence also at

$\theta = m'$. Use the three parts of Lemma 2 in order:

$$\begin{aligned}
&\sum_{n\in[N]} \mathrm{H}(A_n \mid Q_n = q_n, W^{[M]\setminus J}, \theta = m) \\
&= \mathrm{H}(A_{[N]} \mid Q_{[N]} = \mathbf{q}, W^{[M]\setminus J}, \theta = m) \\
&= L + \mathrm{H}(A_{[N]} \mid Q_{[N]} = \mathbf{q}, \\
&\qquad W^{[M]\setminus(J\setminus\{m\})}, \theta = m) \\
&= L + \mathrm{H}(A_{\mathcal{T}} \mid Q_{\mathcal{T}} = q_{\mathcal{T}}, \\
&\qquad W^{[M]\setminus(J\setminus\{m\})}, \theta = m) \\
&= L + \mathrm{H}(A_{\mathcal{T}} \mid Q_{\mathcal{T}} = q'_{\mathcal{T}}, \\
&\qquad W^{[M]\setminus(J\setminus\{m\})}, \theta = m') \\
&= L + \sum_{n\in\mathcal{T}} \mathrm{H}(A_n \mid Q_n = q'_n, \\
&\qquad W^{[M]\setminus(J\setminus\{m\})}, \theta = m') \\
&= L + T\mathrm{H}(A_{n'} \mid Q_{n'} = q'_{n'}, \\
&\qquad W^{[M]\setminus(J\setminus\{m\})}, \theta = m'). \qquad (27)
\end{aligned}$$

The equalities use, successively: conditional independence ($m \in J$); exact recovery of W^m; spanning for the set $J\setminus\{m\}$ (which does not contain the desired index m); T-privacy together with query locality; conditional independence for $m' \in J \setminus \{m\}$; and finally the querywise server symmetry of Lemma 3. □

Proof of Theorem 2. Fix $1 \le s \le M-1$ and any $J \subseteq [M]$ of size s. For every $m \in J$, every supported $\mathbf{q}$ under $\theta = m$, and every server n, we prove by induction on s that

$$\begin{aligned}
&\mathrm{H}(A_n \mid Q_n = q_n, W^{[M]\setminus J}, \theta = m) \\
&\quad = \frac{L}{N^s} \sum_{j=0}^{s-1} N^{s-1-j} T^j. \qquad (28)
\end{aligned}$$

For $s = 1$, take $J = \{m\}$. Since the entire message database is revealed except W^m, the pointwise recovery equality (7) gives

$$\begin{aligned}
&\mathrm{H}(A_{[N]} \mid Q_{[N]} = \mathbf{q}, W^{[M]\setminus\{m\}}, \theta = m) \\
&\quad = L + \mathrm{H}(A_{[N]} \mid Q_{[N]} = \mathbf{q}, W^{[M]}, \theta = m) = L,
\end{aligned}$$

where the final entropy is zero by determinism. By (5) and Lemma 3, the N local entropies are equal and add up to L. Hence each one equals L/N, proving the base case.

Suppose $2 \le s \le M-1$ and that the formula holds for sets of size $s-1$. Choose $m' \in J \setminus \{m\}$ and use Lemma 4. Its left-hand side has N equal terms by Lemma 3. For the query $\mathbf{q}'$ supplied by the lemma, the induction hypothesis applies to $J\setminus\{m\}$, whose size is $s-1$ and which contains m'. Therefore,

$$\begin{aligned}
&N\mathrm{H}(A_n \mid Q_n = q_n, W^{[M]\setminus J}, \theta = m) \\
&= L + T\frac{L}{N^{s-1}} \sum_{j=0}^{s-2} N^{s-2-j} T^j \\
&= \frac{L}{N^{s-1}} \sum_{j=0}^{s-1} N^{s-1-j} T^j. \qquad (29)
\end{aligned}$$

Dividing by N proves (28).

For each fixed local query, the left-hand side of (28) equals the rank, over $\mathbb{F}_p$, of the local answer matrix restricted to the s unfixed message blocks: these blocks are independent and uniform after conditioning on the other $M-s$ messages. Thus this entropy is an integer. At $s = 1$, this gives $N \mid L$. At $s = M-1$ we have

$$\frac{L}{N^{M-1}} \sum_{j=0}^{M-2} N^{M-2-j} T^j \in \mathbb{Z}. \qquad (30)$$

Since $N \mid L$, the last relation can be written as

$$\begin{aligned}
&\frac{L/N}{(N/\gcd(N,T))^{M-2}} \\
&\quad \times \sum_{j=0}^{M-2} \left(\frac{N}{\gcd(N,T)}\right)^{M-2-j} \left(\frac{T}{\gcd(N,T)}\right)^{j} \in \mathbb{Z}. \qquad (31)
\end{aligned}$$

The integer sum is relatively prime to $(N/\gcd(N,T))^{M-2}$: modulo any prime factor of $N/\gcd(N,T)$ it reduces to $(T/\gcd(N,T))^{M-2}$, which is nonzero. Consequently,

$$N\left(\frac{N}{\gcd(N,T)}\right)^{M-2} \mid L,$$

which is exactly (23). For $M = 2$, this conclusion is simply $N \mid L$, already obtained from the base case.

For sufficiency, Xu and Zhang constructed a linear capacity-achieving T-private PIR scheme at the minimum length in (23) over every finite field of cardinality $p \ge N$ [6]. Direct sums yield every positive multiple over the same field. □

For $p < N$, scalar extension yields a more limited sufficient condition. If $p^s \ge N$, a scheme over $\mathbb{F}_{p^s}$ can be viewed as a scheme over $\mathbb{F}_p$ after expressing each field symbol in an $\mathbb{F}_p$-basis. This proves achievability for positive lengths divisible by

$$s\gcd(N,T)\left(\frac{N}{\gcd(N,T)}\right)^{M-1}.$$

The sufficiency of (23) itself over all smaller fields is not asserted here.

V. Discussion: Local Rank Integrality and the Two Length Spectra

We now explain why the noncolluding length spectrum is a numerical semigroup, whereas the $T \ge 2$ result imposes a single divisibility condition. The distinction is not a failure of the capacity converse at $T = 1$: the averaged entropy recursion survives. What fails is its promotion to a recursion for *integer ranks at each supported query*.

A. Averaged and Pointwise Entropy Recursions

Write F_s for the conditional entropy

$$\begin{aligned}
F_s := \mathrm{H}(A_{[N]} \mid Q_{[N]}, W^{[M]\setminus J}, \theta = m), \\
|J| = s, \quad m \in J. \qquad (32)
\end{aligned}$$

For a capacity-achieving T-private scheme, this quantity depends only on s. Indeed, at $s = 1$, the pointwise recovery

equality (7) gives $F_1 = L$. For $s \geq 2$, choose $m' \in J \setminus \{m\}$ and put $S = [M] \setminus J$. The recovery identity, followed by equality in the restriction, privacy, and Han steps of the converse, yields

$$\begin{aligned} F_s &= L + \mathrm{H}(A_{[N]} \mid Q_{[N]}, W^{S\cup\{m\}}, \theta = m) \\ &= L + \frac{T}{N}\mathrm{H}(A_{[N]} \mid Q_{[N]}, W^{S\cup\{m\}}, \theta = m') \\ &= L + \frac{T}{N}F_{s-1}. \end{aligned} \tag{33}$$

The identification of F_{s-1} is justified by induction, since $m' \in J \setminus \{m\}$. Thus, for $1 \leq s \leq M$,

$$F_s = L\sum_{j=0}^{s-1}\left(\frac{T}{N}\right)^j. \tag{34}$$

Equations (33)– (34) hold for both $T = 1$ and $T \geq 2$. They alone imply no rank divisibility: F_s is an *average* over queries of integers and need not itself be an integer.

For $2 \leq T < N$, the situation is stronger. Let $1 \leq s \leq M-1$, $|J| = s$, $m \in J$, and fix a supported full query $\mathbf{q}$. Partition the local answer matrix according to the message blocks:

$$\begin{aligned} \mathbf{G}_n^{q_n} &= \left[\mathbf{G}_{n,1}^{q_n} \ \cdots \ \mathbf{G}_{n,M}^{q_n}\right], \\ A_n &= \mathbf{G}_{n,J}^{q_n} W^J + \mathbf{G}_{n,[M]\setminus J}^{q_n} W^{[M]\setminus J}. \end{aligned} \tag{35}$$

Since W^J is uniform and independent of the known messages and queries, conditioning on $Q_n = q_n$ and $W^{[M]\setminus J}$ removes the second term. The linear image is uniform on its range, so

$$\begin{aligned} &\mathrm{H}(A_n \mid Q_n = q_n, W^{[M]\setminus J}, \theta = m) \\ &\quad = \mathrm{H}(\mathbf{G}_{n,J}^{q_n} W^J) \\ &\quad = \log_p |\operatorname{im} \mathbf{G}_{n,J}^{q_n}| \\ &\quad = \operatorname{rank}_{\mathbb{F}_p} \mathbf{G}_{n,J}^{q_n} \in \mathbb{Z}_{\geq 0}. \end{aligned} \tag{36}$$

This integrality holds even for $T = 1$. The additional consequence of $T \geq 2$ is Lemma 3, which makes the local conditional ranks equal across servers for each supported query. Moreover, the base case and the one-step recurrence show that their common value is independent of the supported query and of the particular J, m of size s. Denote it by h_s. Then

$$\begin{aligned} &Nh_1 = L, \\ &Nh_s = L + Th_{s-1} \qquad (2 \leq s \leq M-1), \end{aligned} \tag{37}$$

so that

$$\begin{aligned} h_s &= \frac{L}{N^s}\sum_{j=0}^{s-1} N^{s-1-j}T^j \\ &= \operatorname{rank}_{\mathbb{F}_p} \mathbf{G}_{n,J}^{q_n} \in \mathbb{Z}_{\geq 0}. \end{aligned} \tag{38}$$

In particular $h_1 = L/N \in \mathbb{Z}$, hence $N \mid L$. Put $d = \gcd(N, T)$. For $1 \leq s \leq M-1$, rearranging (38) gives

$$h_s = \frac{L/N}{(N/d)^{s-1}}\sum_{j=0}^{s-1}\left(\frac{N}{d}\right)^{s-1-j}\left(\frac{T}{d}\right)^j. \tag{39}$$

For $s \geq 2$, if a prime ℓ divides N/d, then

$$\begin{aligned} &\sum_{j=0}^{s-1}\left(\frac{N}{d}\right)^{s-1-j}\left(\frac{T}{d}\right)^j \\ &\qquad \equiv \left(\frac{T}{d}\right)^{s-1} \not\equiv 0 \pmod{\ell}, \end{aligned} \tag{40}$$

since $\gcd(N/d, T/d) = 1$. Thus the numerator sum is relatively prime to $(N/d)^{s-1}$; this also holds trivially for $s = 1$. By the integrality in (38),

$$\left(\frac{N}{d}\right)^{s-1} \Big| \frac{L}{N} \quad\Longrightarrow\quad N\left(\frac{N}{d}\right)^{s-1} \Big| L. \tag{41}$$

At $s = M-1$ this becomes

$$\gcd(N,T)\left(\frac{N}{\gcd(N,T)}\right)^{M-1} \mid L. \tag{42}$$

When $M = 2$, the $s = 1$ case already gives the conclusion. Thus it is precisely the *querywise integer-rank recursion*, not merely (33), that explains the strong divisibility in colluding PIR.

B. The Noncolluding Congruence Constraint

The formal substitution $T = 1$ into (42) would require $N^{M-1} \mid L$. Yet the true $T = 1$ spectrum includes $L = N-1$, so

$$N^{M-1}\mathbb{Z}_{>0} \subsetneq \mathrm{AM}. \tag{43}$$

This is because the symmetry lemma requires $T \geq 2$: to transfer an equality between two distinct servers, both must lie in a single colluding subset. For $T = 1$ the pointwise rank identity (36) still holds, but (37) need not hold.

The remaining integer constraint can be seen directly in terms of the information in *individual* servers' answers. Fix $m \in [M]$ and a query $\mathbf{q}$ supported given $\theta = m$. Write

$$h(\mathbf{q}) := \mathrm{H}(A_{[N]} \mid Q_{[N]} = \mathbf{q}, W^m, \theta = m) \tag{44}$$

and set

$$\delta(\mathbf{q}) := \sum_{n=1}^{N} \mathrm{I}(W^m; A_n \mid Q_{[N]} = \mathbf{q}, \theta = m). \tag{45}$$

The sum $\delta(\mathbf{q})$ is not generally equal to the information about W^m in the *joint* answer: the servers' answers may carry complementary information. In the linear setting, each summand is an integer rank difference, since

$$\begin{aligned} &\mathrm{I}(W^m; A_n \mid Q_{[N]} = \mathbf{q}, \theta = m) \\ &\quad = \operatorname{rank}_{\mathbb{F}_p} \mathbf{G}_n^{q_n} - \operatorname{rank}_{\mathbb{F}_p} \mathbf{G}_{n,[M]\setminus\{m\}}^{q_n} \in \mathbb{Z}_{\geq 0}. \end{aligned} \tag{46}$$

For $T = 1$, the two parts of Lemma 2, with unknown sets $[M]$ and $[M] \setminus \{m\}$, imply

$$\begin{aligned} &\sum_{n=1}^{N} \mathrm{H}(A_n \mid Q_n = q_n, \theta = m) = L + h(\mathbf{q}), \\ &\mathrm{H}(A_n \mid Q_n = q_n, W^m, \theta = m) = h(\mathbf{q}) \quad (n \in [N]). \end{aligned} \tag{47}$$

Consequently,

$$\begin{aligned}\delta(\mathbf{q}) &= \sum_{n=1}^{N}\Big[\mathrm{H}(A_n \mid Q_n = q_n, \theta = m) \\ &\qquad - \mathrm{H}(A_n \mid Q_n = q_n, W^m, \theta = m)\Big] \\ &= L + h(\mathbf{q}) - Nh(\mathbf{q}) \\ &= L - (N-1)h(\mathbf{q}). \end{aligned} \tag{48}$$

By linearity, $h(\mathbf{q})$ and each term in (45) are nonnegative integers. Thus

$$0 \le \delta(\mathbf{q}) \le L, \qquad \delta(\mathbf{q}) \equiv L \pmod{N-1}. \tag{49}$$

Moreover, by the download equality and capacity,

$$\begin{aligned}\mathbb{E}[h(Q_{[N]}) \mid \theta = m] &= D - L \\ &= L\sum_{j=1}^{M-1} N^{-j} \\ &= \frac{L(1 - N^{-(M-1)})}{N-1}, \end{aligned} \tag{50}$$

whence

$$\mathbb{E}[\delta(Q_{[N]}) \mid \theta = m] = \frac{L}{N^{M-1}}. \tag{51}$$

Let b be the least nonnegative residue of L modulo $N-1$. The congruence and nonnegativity in (49) imply, for each supported query,

$$\delta(\mathbf{q}) = b + (N-1)k(\mathbf{q}), \qquad k(\mathbf{q}) \in \mathbb{Z}_{\ge 0}. \tag{52}$$

Taking expectations and using (51),

$$\begin{aligned}\frac{L}{N^{M-1}} &= b + (N-1)\mathbb{E}[k(Q_{[N]}) \mid \theta = m] \\ &\ge b. \end{aligned} \tag{53}$$

As $N^{M-1} \equiv 1 \pmod{N-1}$, this gives

$$\begin{aligned} a &:= \frac{L - bN^{M-1}}{N-1} \in \mathbb{Z}_{\ge 0}, \\ L &= a(N-1) + bN^{M-1}, \end{aligned} \tag{54}$$

and, more precisely,

$$a = N^{M-1}\mathbb{E}[k(Q_{[N]}) \mid \theta = m]. \tag{55}$$

Here a, b are the coefficients in the representation with $0 \le b < N-1$; (55) does not identify the number of component schemes in a particular construction. This argument explains the numerical semigroup as the compatibility of a *pointwise integer congruence* with an *exact prescribed average*. Achievability still relies on the known $L = N-1$ and $L = N^{M-1}$ constructions and their direct sums.

C. Consequences at the Two Generating Lengths

The same reasoning yields distributional statements that hold regardless of the construction. If $L = N-1$, then (48) and nonnegativity imply $h(\mathbf{q}) \in \{0, 1\}$. Equation (50) therefore forces

$$\begin{aligned}&\Pr(h(Q_{[N]}) = 0 \mid \theta = m) = N^{-(M-1)}, \\ &\Pr(h(Q_{[N]}) = 1 \mid \theta = m) = 1 - N^{-(M-1)}. \end{aligned} \tag{56}$$

On a query with $h(\mathbf{q}) = 1$, we have $\delta(\mathbf{q}) = 0$; all individual-server mutual informations about W^m then vanish, although joint recovery remains possible.

For $N \ge 3$ and $L = N^{M-1}$, the least residue is $b = 1$. Thus $\delta(\mathbf{q}) \ge 1$ at every supported query, whereas (51) gives $\mathbb{E}[\delta] = 1$. Hence

$$\begin{aligned}&\delta(\mathbf{q}) = 1, \\ &h(\mathbf{q}) = \frac{N^{M-1} - 1}{N-1} \qquad \text{for every supported } \mathbf{q}. \end{aligned} \tag{57}$$

Since each individual-server mutual information is a nonnegative integer, exactly one server contributes one unit and the others zero, at each supported query. The identity does not require that the contributing server remain the same across queries. More generally, the congruence argument gives

$$1 \le L < N^{M-1} \quad \Longrightarrow \quad N-1 \mid L. \tag{58}$$

D. Querywise Constancy with Colluding Servers

For comparison, use (45) for $2 \le T < N$. At each supported $\mathbf{q}$ for $\theta = m$, conditional independence with unknown set $[M]$ gives the sum of unconditional local entropies. Pointwise recovery and the spanning property with unknown set $[M] \setminus \{m\}$ give

$$\begin{aligned}&\sum_{n=1}^{N} \mathrm{H}(A_n \mid Q_n = q_n, \theta = m) = L + Th_{M-1}, \\ &\sum_{n=1}^{N} \mathrm{H}(A_n \mid Q_n = q_n, W^m, \theta = m) = Nh_{M-1}. \end{aligned} \tag{59}$$

Subtracting these identities, then substituting (38) with $s = M-1$, yields

$$\begin{aligned}\delta(\mathbf{q}) &= L - (N-T)h_{M-1} \\ &= L - (N-T)\frac{L}{N}\sum_{j=0}^{M-2}\left(\frac{T}{N}\right)^j \\ &= L\left(\frac{T}{N}\right)^{M-1}. \end{aligned} \tag{60}$$

Unlike (51), (60) holds for *each* supported query, not only on average. This conclusion concerns the sum of single-server mutual informations; it does not assert equality of all unconditioned server-answer ranks. For $T = 1$, only the average identity (51) is guaranteed.

In summary, the $T \ge 2$ argument synchronizes conditional local ranks strongly enough to impose a querywise integer-rank recurrence. For $T = 1$, the weaker pointwise congruence

and the capacity-determined expectation leave room for the additional generator $N-1$. This comparison is between the *formal* $T=1$ specialization of the $T \geq 2$ formula and the actual noncolluding spectrum: achievable-length sets at distinct actual privacy levels need not be nested, since their target capacities differ. The significance of asymmetry for short noncolluding PIR is consistent with [2], while the colluding subpacketization threshold is known from [5], [6]. The present discussion interprets these length formulas rather than claiming the underlying thresholds as new.

VI. Conclusion

For noncolluding servers, the set of all positive message lengths of capacity-achieving linear PIR is the numerical semigroup generated by $N-1$ and N^{M-1}, with no restriction on the finite-field size. For $2 \leq T < N$, a pointwise entropy argument recovers the known subpacketization divisibility condition under our query-dependent answer-dimension and entropy-download convention; matching constructions are available when the field cardinality is at least N. The two cases are contrasted in Section V: for $T \geq 2$, querywise integer-rank recursions imply divisibility, whereas for $T=1$ the pointwise congruence and a fixed average permit a two-generator numerical semigroup. Note that the capacity being attained depends on T; one cannot in general identify achievable-length sets for distinct values of T by ordinary set inclusion.

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